

Does a Currency Hedge Neutralize Daily Yen Exposure?

货币对冲能否抵消日元的日度风险敞口？

A Multi-Factor Study of HEWJ versus EWJ Before and After the COVID-19 Break

疫情前后 HEWJ 与 EWJ 的多因子比较研究

Course / 课程	STA302 Final Project - Part 1 / STA302 期末项目第一部分
Prepared / 日期	September 21, 2026 / 2026 年 9 月 21 日
Group / 小组	Haiwen Yi; [add all other members / 补充其他成员]
Data / 数据	2,655 daily observations 2014-02-06 to 2026-07-31

English proposal followed by a Chinese study copy / 英文正式提案及中文组内版本

Official English proposal

Abstract

Currency-hedged exchange-traded funds are designed to reduce exchange-rate exposure, but implementation frictions may prevent a complete hedge. We study 2,655 matched daily observations from 2014-2026 and ask how yen appreciation explains the HEWJ-minus-EWJ return spread, conditional on equity, factor, volatility, and interest-rate controls, and whether that relationship changed after March 11, 2020. A multiple linear regression with a pre-specified period interaction estimates yen slopes of -0.851 before and -0.912 after the transition. The preliminary model explains 70.38% of spread variation, but diagnostics identify heteroskedasticity, serial dependence, heavy tails, possible functional-form error, and influential dates. These findings motivate a theory-preserving final analysis using repeated diagnostics, declared sensitivity checks, Newey-West inference, and chronological out-of-sample evaluation. The study is descriptive rather than causal.

Contribution statement

Member	Proposed contribution
Haiwen Yi	Research design, data construction, R analysis, diagnostics, and first draft
[Member 2]	Literature verification, interpretation, and written revision
[Member 3]	Reproducibility check, poster development, and presentation recording
[Member 4, if applicable]	Final-model sensitivity analysis and presentation review

Names and responsibilities must match the separately submitted Group Teamwork Agreement.

Introduction (400 words)

International equity investors receive returns from the local equity market and from the exchange rate used to translate foreign assets into their home currency. Currency-hedged exchange-traded funds attempt to remove the second component, but hedging may be incomplete because forward contracts are rolled periodically, trading calendars differ, expenses and implementation costs exist, and portfolios are not rebalanced continuously. This project asks: **How strongly does daily yen appreciation explain the return difference between the currency-hedged iShares MSCI Japan ETF (HEWJ) and its unhedged counterpart (EWJ), after controlling for equity, factor, volatility, and interest-rate conditions, and did that relationship change after the World Health Organization characterized COVID-19 as a pandemic?**

The response is the daily log-return spread, $Y_t = 100[\Delta \log(\text{HEWJ}_t) - \Delta \log(\text{EWJ}_t)]$, and $\text{JPYapp}_t = -100\Delta \log(\text{DEXJPUS}_t)$ is positive when the yen appreciates. A mean comparison or simple correlation cannot simultaneously adjust for market conditions, estimate conditional period-specific slopes, and test a change in slope. Multiple linear regression provides adjusted coefficients, confidence intervals, and an interaction test. Under a complete contemporaneous hedge, the yen slope should be near -1. A categorical period indicator and its interaction with yen appreciation estimate pre- and post-pandemic slopes; March 11, 2020 is the pre-specified transition and is omitted.

Related work and research gap. Glen and Jorion (1993) find that forward contracts improved the risk-return performance of international bond and equity portfolios from 1974-1990, establishing why hedging can matter without measuring a modern ETF's realized daily offset. Across 17 OECD economies, Hau and Rey (2006) find exchange rates, equity returns, and portfolio flows move jointly, motivating equity-market controls. Campbell, Serfaty-de Medeiros, and Viceira (2010) show with 1975-2005 international asset returns that risk-minimizing currency positions vary across assets and investors, so a one-for-one hedge need not hold. Fama and French (2015) show in diversified U.S. stock portfolios that size, value, profitability, and investment factors explain average-return variation, supporting controls for residual equity composition. In three U.S.-listed currency-hedged ETFs during 2011-2015, Shank and Vianna (2016) find dynamic links among exchange rates, fund trading, and benchmarks. None of these studies estimates the realized daily yen exposure of the matched HEWJ/EWJ pair or tests a pre-specified 2020 slope change. This project fills that gap with an auditable daily matched-pair design.

The results would benefit U.S.-dollar investors, portfolio managers, and risk teams deciding whether a “hedged” Japan allocation actually offsets daily yen exposure. The pandemic indicator is descriptive, not causal, because many policies and market conditions changed simultaneously.

Data description (289 words)

Variable	Type	Mean	SD	Range	Missing	Important feature
Response spread	Numerical	0.018	0.624	-4.310 to 3.783	0	Extreme days; heavy tails
Yen appreciation	Numerical	-0.010	0.581	-2.850 to 4.322	0	Wide daily extremes
Nikkei return	Numerical	0.033	1.358	-13.234 to 9.737	0	Large crash/rebound days
Japan SMB	Numerical	-0.004	0.500	-2.340 to 3.120	0	Moderate spread
Japan HML	Numerical	0.018	0.727	-3.590 to 4.570	0	Relatively wide factor range
Japan RMW	Numerical	-0.003	0.418	-1.840 to 2.220	0	Moderate spread
Japan CMA	Numerical	0.009	0.389	-2.370 to 2.290	0	Moderate spread
Japan MOM	Numerical	-0.009	0.813	-7.150 to 8.550	0	Wide factor extremes
VIX log change	Numerical	-0.221	8.053	-44.245 to 76.825	0	Largest dispersion and right extreme
Lagged rate difference	Numerical	1.846	1.825	-0.010 to 5.394	0	Slow-moving monthly series
Post period	Categorical	Pre: 1,304	Post: 1,351	2 levels	0	WHO-defined break

Table 1. Summary and missingness for all preliminary-model variables.

The analysis contains 2,655 exact-interval daily observations from February 6, 2014 through July 31, 2026. Yahoo Finance records traded HEWJ and EWJ quotations for market information; adjusted closes incorporate splits and distributions for total-return comparisons. The Federal Reserve Board collects DEXJPUS as the noon New York buying rate for yen transfers. The Nikkei Industry Research Institute records the daily close of 225 liquid Tokyo stocks; CBOE derives VIX from index-option prices as expected near-term volatility; and the OECD compiles monthly U.S. and Japanese call-money/interbank rates. FRED redistributes these series. Kenneth French's Data Library forms value-weighted Japan equity portfolios for asset-pricing research, producing daily five-factor and momentum returns.

The daily HEWJ-minus-EWJ log-return spread has mean 0.0180 percentage points, standard deviation 0.6243, and range -4.3097 to 3.7826. It is continuous and signed, making a linear conditional mean interpretable. Daily observations are ordered rather than strictly independent, so residual dependence is assessed and no causal interpretation is made.

Table 1 summarizes every model variable and missingness. The analysis sample has zero missing values because rows are retained only when required sources share the same return interval; this is an inclusion rule, not imputation. Extreme movements appear in the response, Nikkei return, momentum, and VIX change. Monthly rates are carried forward with a one-month lag. Post contains 1,304 pre-period and 1,351 post-period observations. Price returns are computed on native calendars before merging.

The model uses ten predictors: yen appreciation, Post, Nikkei return, SMB, HML, RMW, CMA, MOM, log VIX change, and the lagged U.S.-Japan rate differential. The JPYapp $\times$ Post interaction answers the research question and reflects prior evidence of time-varying hedge dynamics; Figure 1 shows its period-specific slopes. The split follows the WHO announcement rather than a search for the smallest p-value.

Ethics discussion (198 words)

The dataset is trustworthy because its provenance, ownership, and transformations are documented. Yahoo Finance closes, FRED series, and Kenneth French factors are public sources, but authority does not eliminate error: providers can revise observations, calendars may differ, and adjusted prices depend on provider methods. We preserve snapshots and SHA-256 checksums, document transformations, and avoid overstating precision. This addresses the module's trustworthiness criteria of provenance, transparency, accuracy, and reproducibility.

Collection and use are ethically low risk. The sources contain aggregate market prices and factor portfolios, not personal information, private holdings, individual trading records, or human-subject data. Consent, confidentiality, and re-identification risks are minimal. We respect ownership by citing each provider, limiting source-file access to the course group and graders, and not publicly redistributing restricted Nikkei observations or claiming ownership of third-party data.

The principal ethical risk is interpretation. A descriptive association could be presented incorrectly as investment advice or evidence that COVID-19 caused a change. We will state that results apply to one ETF pair, acknowledge survivorship and product-selection limitations, and avoid causal language because changes were uncontrolled. Historical predictive performance is not a promise of future returns. These limits make the use proportionate to the data and question.

Preliminary results (391 words including captions)

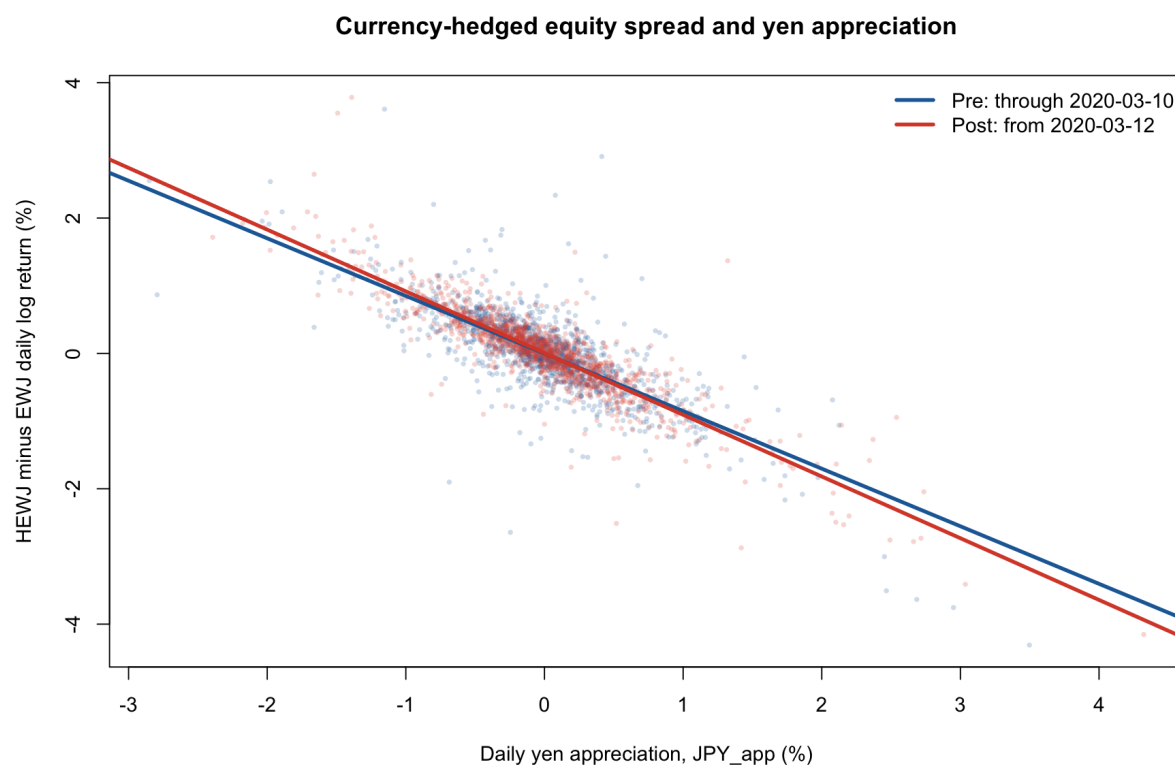


Figure 1. Daily return spread versus yen appreciation with period-specific controlled OLS slopes.

Term	Estimate	Classical SE	95% CI	p
Intercept	-0.0010	0.0102	[-0.0210, 0.0191]	0.924
Yen appreciation	-0.8509	0.0184	[-0.8870, -0.8148]	<0.001
Post indicator	0.0072	0.0151	[-0.0225, 0.0368]	0.636
Nikkei return	-0.0034	0.0061	[-0.0155, 0.0086]	0.575
SMB	0.0199	0.0154	[-0.0102, 0.0501]	0.195
HML	-0.0235	0.0172	[-0.0572, 0.0101]	0.171
RMW	-0.0114	0.0274	[-0.0650, 0.0423]	0.677
CMA	0.0381	0.0245	[-0.0100, 0.0862]	0.121
MOM	0.0017	0.0088	[-0.0156, 0.0190]	0.845
VIX log change	-0.0068	0.0008	[-0.0085, -0.0051]	<0.001

Term	Estimate	Classical SE	95% CI	p
Lagged rate difference	0.0027	0.0041	[-0.0054, 0.0108]	0.510
Yen x Post	-0.0612	0.0235	[-0.1073, -0.0151]	0.009

Table 2. Preliminary OLS estimates with classical standard errors and 95% confidence intervals.

The full ordinary least-squares model is

$$Y(t) = b_0 + b_1 JPYapp(t) + b_2 Post(t) + b_3 [JPYapp(t) \times Post(t)] + g'Z(t) + e(t),$$

where Z_t contains the eight market and factor controls. Table 2 reports every coefficient with a classical standard error and 95% confidence interval. The model explains 70.38% of daily spread variation (adjusted $R^2=70.26\%$). The pre-period yen slope is -0.851 (SE 0.0184; 95% CI [-0.887, -0.815]). Thus, conditional on the controls, a one-percentage-point yen appreciation is associated with a 0.851-percentage-point lower HEWJ-minus-EWJ return. The interaction is -0.0612 (SE 0.0235; $p = 0.0093$), giving an implied post-period slope of -0.912. The period level shift is not significant. VIX log change is negative; most other controls are imprecisely estimated.

The strong negative currency slope is consistent with Hau and Rey's joint currency-equity mechanism and with the hedge interpretation. Its departure from exactly -1 agrees with Campbell et al.'s result that effective hedges need not be mechanical one-for-one positions. The mostly small factor coefficients suggest that pairing HEWJ with EWJ removes much common equity exposure, while retaining Fama-French controls prevents that conclusion from being assumed. The changed slope is directionally compatible with Shank and Vianna's evidence of time-varying currency-hedged ETF dynamics, although our pandemic comparison is not causal.

Figure 2 provides a six-panel preliminary diagnostic grid. Residuals versus fitted values show no dominant smooth curve, but the RESET test rejects exact functional form ($p < 0.001$), so linearity remains questionable. The scale-location panel and Breusch-Pagan test indicate nonconstant variance ($p = 0.0082$). The Q-Q plot has heavy tails and Jarque-Bera rejects normality ($p < 0.001$). Residuals over time and the ACF show dependence; the five-lag Breusch-Godfrey test rejects independence ($p < 0.001$), and Durbin-Watson is 2.83. The maximum VIF is 3.56, so severe multicollinearity is not evident. The Cook's-distance panel identifies 134 cases above the screening threshold $4/n$, with maximum 0.142; these observations are flagged for sensitivity analysis rather than automatically deleted.

These are preliminary OLS results. In accordance with the Part 1 instruction, we diagnose but do not correct violations here; robust inference and sensitivity analysis are deferred to the final project.

Plan for the remaining analysis (285 words)

The focal terms - yen appreciation, Post, and their interaction - will remain because they define the research question. Other controls will first be retained on substantive grounds. We will compare theory-preserving nested models using partial F-tests, adjusted R^2 , residual behavior, and focal-coefficient stability. We will not drop a control solely because its p-value exceeds 0.05, and hierarchy will be respected.

After every material specification change, we will recheck residual-versus-fitted, Q-Q, time-order, and ACF plots, Breusch-Pagan and Breusch-Godfrey tests, Cook's distance, and VIF. Because the model shows heteroskedasticity, dependence, heavy tails, and functional-form evidence, we will examine a pre-specified nonlinear yen term and a Yeo-Johnson response transformation, which is valid for signed returns; a direct logarithm is invalid. Transformations will be compared for assumption improvement and interpretability, not selected to manufacture significance.

Influential dates will be documented before any sensitivity run. The primary analysis will retain all observations; secondary analyses will compare declared 0.5/99.5-percentile winsorization and the exclusion of individually justified data errors, if any. Once a final functional form is chosen, classical intervals will be supplemented by Newey-West intervals with 1-, 5-, and 10-day lags. We will also vary the pandemic cut within a small declared window rather than search all dates.

Prediction remains secondary. A fixed chronological 60/20/20 split will fit candidates on earlier training dates, select by tuning RMSE, and evaluate the locked choice once on the untouched test period against historical-mean and zero-return benchmarks. No random cross-validation or test-set tuning will be used. The frozen raw files, cleaned CSV, code, and outputs will be versioned together.

The schedule in Table 3 assigns analysis, poster, and recording milestones. Member placeholders will be replaced with names from the signed Teamwork Agreement before submission.

Dates	Milestone	Owner(s)	Checkable output
Nov 9-15	Audit cleaned data and diagnostics	Haiwen + Member 3	Frozen data; issue log
Nov 16-22	Revise model and run sensitivities	Haiwen + Member 2	Final model tables/figures
Nov 23-27	Create and review poster	Member 2 + Member 3	Complete poster draft
Nov 28-30	Record, caption, and review	All members	Final recording
Dec 1-2	Final file and link audit	Haiwen + all members	Submission-ready package

Table 3. Proposed team schedule. Member placeholders must be replaced with the signed roster.

References

- Campbell, J. Y., Serfaty-de Medeiros, K., & Viceira, L. M. (2010). Global currency hedging. *The Journal of Finance*, 65(1), 87-121. <https://doi.org/10.1111/j.1540-6261.2009.01524.x>
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- Shank, C. A., & Vianna, A. C. (2016). Are US-dollar-hedged-ETF investors aggressive on exchange rates? A panel VAR approach. *Research in International Business and Finance*, 38, 430-438. <https://doi.org/10.1016/j.ribaf.2016.05.002>

Data and product documentation

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World Health Organization. (2020, March 11). *WHO Director-General's opening remarks at the media briefing on COVID-19*. <https://www.who.int/>

Yahoo Finance. (n.d.). *HEWJ and EWJ historical data* [Data sets]. <https://finance.yahoo.com/>

Submission items requiring group confirmation

1. Replace bracketed member names and role assignments with the actual group roster.
2. Complete and sign the course's Group Teamwork Agreement PDF using the same names and responsibilities.
3. Upload the prepared original and cleaned CSV folders to UofT OneDrive and add a Quercus submission comment with permission set to "anyone in UofT with the link can access."

Residual diagnostics

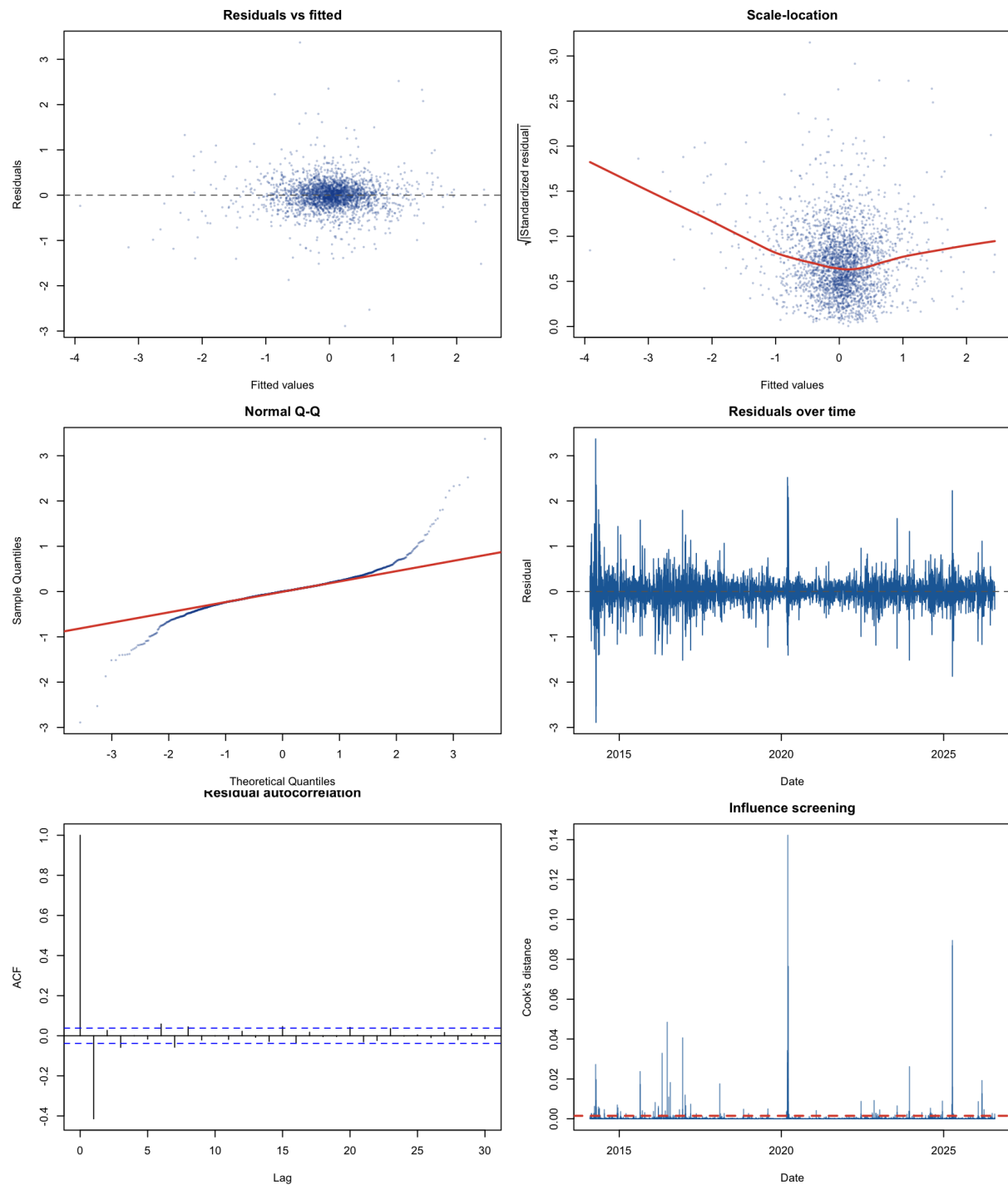


Figure 2. Six-panel diagnostics for the uncorrected preliminary OLS model.

第二部分 - 中文提案（组内讨论与理解版本）

正式课程提交请使用独立英文 PDF；本部分与英文版采用相同研究设计、数值与限制。

摘要

货币对冲 ETF 的目标是降低汇率风险，但展期、费用、交易日差异和非连续再平衡可能造成不完全对冲。本研究使用 2014-2026 年 2,655 个严格同期日度观测，检验日元升值在控制股票市场、资产定价因子、波动率和利率条件后如何解释 HEWJ-EWJ 收益差，并比较 2020 年 3 月 11 日前后的条件斜率。预先设定时期交互项的多元线性回归得到疫情前 -0.851、疫情后 -0.912 的日元斜率，初步模型解释 70.38% 的收益差变异。诊断同时发现异方差、序列依赖、厚尾、可能的函数形式问题和影响点，因此最终分析将采用重复诊断、预先声明的敏感性检验、Newey-West 推断和严格的时间顺序样本外评估。本研究只作描述性解释，不作因果推断。

成员贡献说明

成员	计划贡献
Haiwen Yi	研究设计、数据构建、R 分析、诊断检验和初稿
[成员 2]	文献核验、结果解释和文字修订
[成员 3]	可复现性检查、海报制作和演示录制
[成员 4，如适用]	最终模型敏感性分析与演示审核

姓名与职责必须和单独提交的小组协议一致。

研究背景与问题

国际股票投资者的收益同时受到本地股票市场和汇率影响。货币对冲 ETF 试图消除汇率部分，但由于远期合约展期、交易日差异、费用和非连续再平衡，对冲可能不完整。本项目研究：在控制股票市场、资产定价因子、波动率和利率条件后，日元升值在多大程度上解释货币对冲型日本股票 ETF（HEWJ）与非对冲型 ETF（EWJ）的日度收益差？这种关系在世界卫生组织宣布 COVID-19 为全球大流行后是否改变？

响应变量是日度对数收益差 $Y_t = 100[\Delta \log(\text{HEWJ}_t) - \Delta \log(\text{EWJ}_t)]$ ，核心预测变量为 $\text{JPYapp}_t = -100\Delta \log(\text{DEXJPUS}_t)$ 。简单均值比较或相关系数无法同时控制市场条件、估计分时期条件斜率并检验斜率是否改变；多元线性回归可以通过系数、置信区间和交互项检验回答这些问题。完整同期对冲下，日元斜率应接近 -1。模型以分类变量 Post 及其与日元升值的交互项估计疫情前后斜率，并剔除转换日 2020 年 3 月 11 日。

相关研究与研究缺口。Glen 与 Jorion（1993）发现，1974-1990 年国际股票和债券组合加入远期合约后风险收益表现改善，说明对冲有价值，但没有衡量现代 ETF 的日度实际抵消程度。Hau 与 Rey（2006）在 17 个 OECD 经济体中发现汇率、股票收益与资金流动共同变化。Campbell 等（2010）利用 1975-2005 年国际资产收益说明风险最小化货币头寸随资产和投资者而异。Fama 与 French（2015）在美国多元股票组合中发现规模、价值、盈利和投资因子解释平均收益差异。Shank 与 Vianna（2016）在 2011-2015 年三个美国上市货币对冲 ETF 中发现汇率、交易与基准之间存在动态联系。现有研究没有直接估计 HEWJ/EWJ 配对的日度实际日元敞口，也没有检验预先设定的 2020 年斜率变化。本研究以可审计的日度配对设计填补这一具体缺口。结果可帮助美元投资者、组合经理和风险团队；疫情指标只描述结构变化，不代表因果效应。

数据说明

变量	类型	均值	标准差	范围	缺失	主要特征
响应变量收益差	数值	0.018	0.624	-4.310 to 3.783	0	存在极端日与厚尾
日元升值	数值	-0.010	0.581	-2.850 to 4.322	0	日度极端值较宽
日经收益	数值	0.033	1.358	-13.234 to 9.737	0	存在大跌与反弹日
日本 SMB	数值	-0.004	0.500	-2.340 to 3.120	0	离散程度中等
日本 HML	数值	0.018	0.727	-3.590 to 4.570	0	因子范围较宽
日本 RMW	数值	-0.003	0.418	-1.840 to 2.220	0	离散程度中等
日本 CMA	数值	0.009	0.389	-2.370 to 2.290	0	离散程度中等
日本 MOM	数值	-0.009	0.813	-7.150 to 8.550	0	存在较宽极端值
VIX 对数变化	数值	-0.221	8.053	-44.245 to 76.825	0	离散最大且右侧极端
滞后利差	数值	1.846	1.825	-0.010 to 5.394	0	按月更新、变化缓慢
疫情后时期	分类	疫情前：1.3 04	疫情后：1.3 51	2 个水平	0	WHO 事件定义断点

表 1：全部初步模型变量的数值汇总与缺失数量。

分析包含 2014 年 2 月 6 日至 2026 年 7 月 31 日的 2.655 个严格同期日度观测。Yahoo Finance 记录 HEWJ 和 EWJ 的市场报价；复权收盘价包含拆股与现金或资本利得分配调整。美国联邦储备委员会采集纽约中午日元兑美元买入价；Nikkei Industry Research Institute 记录东京证券交易所 225 只高流动性股票的日收盘指数；CBOE 用指数期权价格构造 VIX 以衡量近期预期波动；OECD 汇编美日月度同业拆借/通知贷款利率。FRED 负责再发布这些序列。Kenneth French Data Library 为资产定价研究将日本股票构造成价值加权组合，并据此生成日度五因子和动量收益。

响应变量均值为 0.0180 个百分点、标准差 0.6243，范围为 -4.3097 至 3.7826。它是可正可负的连续变量，因此条件均值线性模型具有可解释性；但日度观测并非严格独立，所以残差分析会正式检验时间依赖。表 1 汇总全部模型变量及缺失数量。严格同期样本没有缺失值，因为只有所需来源共享同一收益区间时才保留该行；这是预先声明的纳入规则，不是插补。收益差、日经收益、动量和 VIX 变化包含极端日；月度利率滞后一个月后向前填充。Post 含疫情前 1,304 个和疫情后 1,351 个观测，各价格序列先在自身交易日历内计算收益。

模型包含十个预测变量：日元升值、Post、日经收益、SMB、HML、RMW、CMA、MOM、VIX 对数变化及滞后美日利差。JPYapp × Post 直接回答研究问题，也符合既有研究中对冲效果随时间变化的证据；图 1 展示分时期斜率。断点来自 WHO 事件，而不是事后搜索最小 p 值。

伦理声明

数据具有可追溯的来源、所有权和变换记录，因此基本可信。Yahoo Finance、FRED 与 Kenneth French 是公开来源，但权威来源仍可能修订观测，交易日历也可能不一致，复权价格还依赖提供方法。项目保存冻结快照与 SHA-256 校验值、记录全部变换并避免夸大精度，对应伦理模块中的来源可追溯、透明度、准确性和可复现性标准。

数据只包含聚合市场价格和因子组合，不包含个人信息、私人持仓、个人交易或人体研究记录，因此同意、保密和再识别风险很低。项目会引用每个提供方，把源文件访问限制在课程小组和评分人员，不公开再分发受限制的 Nikkei 观测，也不宣称拥有第三方数据。

主要伦理风险来自解释。描述性关联不应被写成投资建议或 COVID-19 的因果效应。报告会说明结论仅适用于一个 ETF 配对，承认存续偏差和产品选择限制，并避免因果措辞。历史预测表现也不代表未来收益承诺。

初步结果

变量	估计值	经典标准误	95% 区间	p
截距	-0.0010	0.0102	[-0.0210, 0.0191]	0.924
日元升值	-0.8509	0.0184	[-0.8870, -0.8148]	<0.001
疫情后指标	0.0072	0.0151	[-0.0225, 0.0368]	0.636
日经收益	-0.0034	0.0061	[-0.0155, 0.0086]	0.575
SMB	0.0199	0.0154	[-0.0102, 0.0501]	0.195
HML	-0.0235	0.0172	[-0.0572, 0.0101]	0.171
RMW	-0.0114	0.0274	[-0.0650, 0.0423]	0.677
CMA	0.0381	0.0245	[-0.0100, 0.0862]	0.121
MOM	0.0017	0.0088	[-0.0156, 0.0190]	0.845
VIX 对数变化	-0.0068	0.0008	[-0.0085, -0.0051]	<0.001
滞后利差	0.0027	0.0041	[-0.0054, 0.0108]	0.510
日元 x 疫情后	-0.0612	0.0235	[-0.1073, -0.0151]	0.009

表 2：完整初步 OLS 系数、经典标准误与 95% 置信区间。

完整普通最小二乘模型为：

$$Y(t) = b_0 + b_1 \text{JPYapp}(t) + b_2 \text{Post}(t) + b_3 [\text{JPYapp}(t) \times \text{Post}(t)] + g'Z(t) + e(t).$$

表 2 给出全部系数、经典标准误和 95% 置信区间。模型解释 70.38% 的日度收益差变异（调整后 $R^2=70.26\%$ ）。疫情前日元斜率为 -0.851（SE 0.0184；95% CI [-0.887, -0.815]）；交互项为 -0.0612（SE 0.0235； $p = 0.0093$ ），得到疫情后隐含斜率 -0.912。时期水平变化不显著，VIX 对数变化系数为负，其他多数控制项估计不精确。

强负斜率符合 Hau 与 Rey 的汇率-股票联动机制；斜率并非严格 -1 与 Campbell 等人的非机械式最优对冲结果一致。多数因子系数较小，说明配对差分消除了大量共同股票风险。斜率变化方向上与 Shank 和 Vianna 的动态关系结果相容，但不能解释为疫情因果效应。

图 2 给出六面板初步诊断。残差-拟合值图没有明显单一曲线，但 RESET 拒绝精确函数形式（ $p < 0.001$ ）；尺度-位置图和 Breusch-Pagan 检验说明方差不恒定（ $p = 0.0082$ ）；Q-Q 图厚尾且 Jarque-Bera 拒绝正态性；时间残差图和 ACF 显示依赖，五阶 Breusch-Godfrey 拒绝独立性（ $p < 0.001$ ），Durbin-Watson 为 2.83。最大 VIF 为 3.56。Cook 距离面板识别出 134 个超过筛查阈值 $4/n$ 的观测，最大值 0.142；这些日期只被标记用于敏感性分析，不会自动删除。

本部分只诊断、不修正模型违背；稳健推断和敏感性分析留到最终项目。

后续分析计划

日元升值、Post 和交互项定义研究问题，因此必须保留。其他市场、因子、波动率和利率控制先依据理论保留，再通过部分 F 检验、调整后 R^2 、残差表现和核心系数稳定性比较保持理论含义的嵌套模型。不会仅因单个 p 值大于 0.05 而删除变量，并始终遵守交互项层级原则。

每次实质性修改后都会重新检查残差-拟合值图、Q-Q 图、时间图、ACF、Breusch-Pagan、Breusch-Godfrey、Cook 距离和 VIF。针对异方差、序列相关、厚尾和函数形式证据，将预先比较非线性日元项和可处理负值的 Yeo-Johnson 响应变换；不能直接对可为负响应变量取对数。变换依据假设改善和可解释性选择，而不是为了制造显著性。

主分析保留所有观测；次要分析才使用预先声明的 0.5/99.5 分位缩尾或删除经证实的数据错误。最终函数形式确定后再加入 1、5、10 日 Newey-West 区间，并在小范围事先声明窗口内改变疫情断点。预测只作为次要检查，采用固定的 60/20/20 时间切分：训练集拟合、调优集按 RMSE 选模、测试集只做一次最终评估，并与历史均值和零收益基准比较；不做随机交叉验证，也不使用测试集调参。

表 3 安排分析、海报和录制里程碑。提交前需把成员占位符替换成签署协议中的真实姓名。

日期	里程碑	负责人	可检查输出
11 月 9-15 日	复核清理数据与诊断	Haiwen + 成员 3	冻结数据、问题日志
11 月 16-22 日	模型修订与敏感性分析	Haiwen + 成员 2	最终模型表与图
11 月 23-27 日	制作并审核海报	成员 2 + 成员 3	完整海报草稿
11 月 28-30 日	录制、字幕与复核	全组	最终视频
12 月 1-2 日	文件与链接最终检查	Haiwen + 全组	可提交包

表 3：拟定团队进度。提交前必须替换成员占位符。

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芝加哥期权交易所（无日期）。CBOE 波动率指数 VIX [VIXCLS]。FRED，圣路易斯联邦储备银行。 <https://fred.stlouisfed.org/series/VIXCLS>

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Kenneth R. French Data Library（无日期）。日本五因子（日度）与日本动量因子（日度）。 https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Nikkei Industry Research Institute（无日期）。日经平均指数 Nikkei 225 [NIKKEI225]。FRED，圣路易斯联邦储备银行。 <https://fred.stlouisfed.org/series/NIKKEI225>

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提交前仍需确认

1. 用真实小组名单替换成员和职责占位符。
2. 完成并签署课程 Group Teamwork Agreement PDF，确保姓名与职责一致。
3. 把准备好的原始和清理 CSV 文件夹上传至 UofT OneDrive，并在 Quercus 评论中加入链接，权限设为 “anyone in UofT with the link can access”。